Dispersion and the Meaning of the Wave Function

PHYS 2203 — Introductory Physics III

November 4, 2026

Overview

Last lecture showed that a free particle's wave packet moves at the group velocity and spreads as it moves. We begin by recapping that result and identifying what controls it: the speed of the packet comes from the slope of $\omega(k)$ at the packet's central wave number, and the rate of spreading from its curvature. This leads to the dispersion relation $\omega(k)$, which we compare for light and for matter waves. We close by looking carefully at how the wave function gives the probability of finding a particle, and how to calculate that probability for any region.

1 Recap: A Moving Wave Packet

Last lecture we let a free wave packet evolve in time by giving each of its plane waves its own time dependence:

$$\Psi(x,t) = \int_{-\infty}^{+\infty} A(k)\, e^{i(kx-\omega(k)t)}\, \mathrm{d}k, \qquad \omega(k) = \frac{\hbar k^2}{2m}. \tag{1}$$

For a Gaussian packet of initial width ε, built from wave numbers centered on k_0, the integral can be done exactly, and the probability density is

$$|\Psi(x,t)|^2 = \frac{C^2}{\sqrt{1+\hbar^2t^2/4m^2\varepsilon^4}} \exp\left[\frac{-(x-v_0t)^2}{2\varepsilon^2\left(1+\hbar^2t^2/4m^2\varepsilon^4\right)}\right]. \tag{2}$$

Two quantities control how the packet behaves.

* **The speed of the packet, v_0.** The peak is at $x = v_0t$, where

$$v_0 = \frac{\hbar k_0}{m} = \frac{p_0}{m}. \tag{3}$$

  This is the velocity of a classical particle whose momentum is the packet's central momentum $p_0 = \hbar k_0$. It is *not* the speed $\omega_0/k_0 = \hbar k_0/2m = v_0/2$ of the crests of the central plane wave.

* **The rate of spreading, set by $\hbar/m$.** The width grows from ε to $\varepsilon\sqrt{1+(\hbar/m)^2t^2/4\varepsilon^4}$.

Both quantities come from derivatives of $\omega(k)$ at the center of the packet:

$$\left.\frac{\mathrm{d}\omega}{\mathrm{d}k}\right|_{k_0} = \frac{\hbar k_0}{m} = v_0, \qquad \left.\frac{\mathrm{d}^2\omega}{\mathrm{d}k^2}\right|_{k_0} = \frac{\hbar}{m}. \tag{4}$$

This is true for any relation $\omega(k)$. A Gaussian packet built from wave numbers near k_0 has

A moving Gaussian packet.

$$|\Psi(x,t)|^2 = \frac{C^2}{\sqrt{1+D^2t^2/4\varepsilon^4}}\exp\left[\frac{-(x-v_{\text{group}}t)^2}{2\varepsilon^2\,(1+D^2t^2/4\varepsilon^4)}\right], \qquad v_{\text{group}} = \left.\frac{\mathrm{d}\omega}{\mathrm{d}k}\right|_{k_0}, \quad D = \left.\frac{\mathrm{d}^2\omega}{\mathrm{d}k^2}\right|_{k_0}.$$

The **slope** of $\omega(k)$ at k_0 sets the speed of the packet, the group velocity. The **curvature** D sets how quickly it spreads. If $D = 0$, the packet slides along without changing shape. Both are evaluated at k_0 because the packet is built mostly from wave numbers near k_0, where $A(k)$ is largest.

2 Dispersion Relations

The relation $\omega(k)$ is called the **dispersion relation**. With $E = \hbar\omega$ and $p = \hbar k$, it is the wave version of the relation between a particle's energy and momentum. Everything about how a packet moves follows from it: the phase velocity ω/k, the group velocity $\mathrm{d}\omega/\mathrm{d}k$, and the spreading $D = \mathrm{d}^2\omega/\mathrm{d}k^2$.

	$\omega(k)$	$v_{\text{phase}} = \omega/k$	$v_{\text{group}} = \mathrm{d}\omega/\mathrm{d}k\vert_{k_0}$	$D = \mathrm{d}^2\omega/\mathrm{d}k^2\vert_{k_0}$
Light ($E = pc$)	ck	c	c	0
Matter ($E = p^2/2m$)	$\dfrac{\hbar k^2}{2m}$	$\dfrac{\hbar k_0}{2m} = \dfrac{v_0}{2}$	$\dfrac{\hbar k_0}{m} = v_0$	$\dfrac{\hbar}{m}$

- **Light in vacuum.** $\omega(k)$ is a straight line, so every plane wave travels at c and $D = 0$. A light pulse moves at c and keeps its shape.
- **Matter waves.** The group velocity is v_0, the velocity of the particle. The phase velocity is only half of that, but no measurement depends on it, since it disappears from $|\Psi|^2$. This resolves the puzzle of Lecture 25, where $f\lambda = E/p$ was not the particle's speed. Because $\omega(k)$ is curved, $D \neq 0$ and a matter-wave packet spreads, even in empty space.
- **Graphically,** v_{phase} is the slope of the line from the origin to the point (k_0, ω_0), v_{group} is the slope of the tangent to the curve at k_0, and D measures how much the curve bends there (Figure 1).

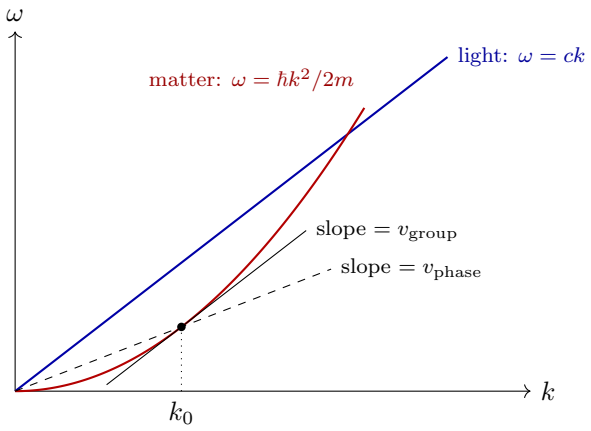


Figure 1: Dispersion relations. For matter waves, $\omega = \hbar k^2/2m$ is a parabola. At k_0 the phase velocity is the slope of the line from the origin (dashed), and the group velocity is the slope of the tangent (solid), twice as large.

3 Interpreting the Wave Function

The wave function itself is never measured. What it tells us is the probability of finding the particle in a given place.

A probability density. $|\Psi(x,t)|^2$ is a probability *per unit length*, with units of m^{-1}, much like the mass per unit length of a rod whose density varies along its length. To find the mass of a piece of the rod, we integrate the density over that piece. In the same way, the probability of finding the particle between $x = a$ and $x = b$ is the area under $|\Psi|^2$ between those points:

Probability in a region.

$$P(a < x < b) = \int_a^b |\Psi(x,t)|^2 \, \mathrm{d}x.$$

* **Small regions.** If $|\Psi|^2$ is nearly constant over a short interval Δx, the integral is approximately $|\Psi(x)|^2 \, \Delta x$. The probability of finding the particle at one exact point is zero. Only regions have probabilities.

* **Normalization.** The particle must be found somewhere, so the total area is 1:

$$\int_{-\infty}^{+\infty} |\Psi(x,t)|^2 \, \mathrm{d}x = 1 \tag{5}$$

  This condition fixes the overall constant in front of a wave function. A density can be larger than 1 (for example, 5 nm^{-1}), but a probability cannot.

* **Phases drop out.** Multiplying Ψ by a factor of magnitude 1, such as $e^{ik_0 x}$ or $e^{-i\omega t}$, does not change $|\Psi|^2$, and so does not change where the particle is likely to be found.

* **Many measurements.** A single measurement finds the particle at one definite place that cannot be predicted. If the same wave function is prepared many times and the position

measured each time, the fraction of results in a region approaches $P(a < x < b)$, and a histogram of the results takes the shape of $|\Psi|^2$.

Example. A particle confined between $x = 0$ and $x = L$ has the wave function $\psi(x) = Ax(L-x)$, and $\psi = 0$ outside. To normalize, expand the integrand and integrate term by term:

$$\int_0^L x^2(L-x)^2\,\mathrm{d}x = \int_0^L \left(L^2x^2 - 2Lx^3 + x^4\right)\mathrm{d}x = \frac{L^5}{3} - \frac{2L^5}{4} + \frac{L^5}{5} = L^5\left(\frac{10-15+6}{30}\right) = \frac{L^5}{30}. \quad (6)$$

Setting $A^2L^5/30 = 1$ gives $A = \sqrt{30/L^5}$.

The probability of finding the particle in the left quarter of the region is the same integral from 0 to $L/4$. Substituting $u = x/L$ (so $x = Lu$ and $\mathrm{d}x = L\,\mathrm{d}u$),

$$P\left(0 < x < \tfrac{L}{4}\right) = \frac{30}{L^5}\int_0^{L/4} x^2(L-x)^2\,\mathrm{d}x = 30\int_0^{1/4} u^2(1-u)^2\,\mathrm{d}u = 30\int_0^{1/4}\left(u^2 - 2u^3 + u^4\right)\mathrm{d}u, \quad (7)$$

$$P = 30\left[\frac{u^3}{3} - \frac{u^4}{2} + \frac{u^5}{5}\right]_0^{1/4} = 30\left(\frac{1}{192} - \frac{1}{512} + \frac{1}{5120}\right) = 30\left(\frac{80-30+3}{15\,360}\right) = 0.10, \quad (8)$$

and by symmetry the same in the right quarter, so the particle is in the middle half 80% of the time (Figure 2). The density at the center is $|\psi(L/2)|^2 = 30/16L = 1.9/L$, so the probability of finding the particle within a small interval $\Delta x = 0.01L$ at the center is about $(1.9/L)(0.01L) = 0.019$.

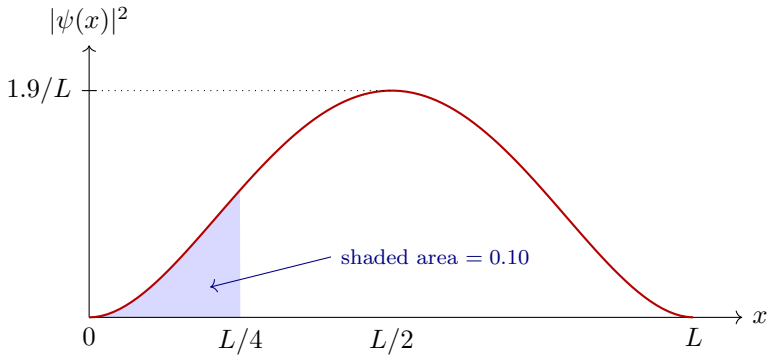


Figure 2: The probability density for $\psi(x) = Ax(L-x)$. The shaded area, 0.10, is the probability of finding the particle between $x = 0$ and $x = L/4$. The total area under the curve is 1.